

Topic 2 Chapter 3 Worksheet

In the worksheets for this class, you are going to create SQL statement for each task described below.

You are required to take a screen shot of the SQL code and the output. Paste the image after each numbered question.

Save your work and upload the document to the assignment folder by the due date in the Syllabus.

If you want, you may delete the sample output images.

Examples of what to turn in:

Display all movies with a content rating of G.

```
15 • SELECT movie_title, content_rating FROM mytable WHERE content_rating = 'G';
```

movie_title	content_rating
Monsters University	G
Cars 2	G
Toy Story 3	G
WALL·E	G

Display all movies with that have the name Leonardo in actor_1_name.

```
15 • SELECT movie_title, actor_1_name as 'Actor 1 with with the name Leonardo' FROM mytable
16 WHERE actor_1_name like '%Leonardo%';
```

movie_title	Actor 1 with with the name Leonardo
Titanic	Leonardo DiCaprio
The Great Gatsby	Leonardo DiCaprio

For your reference, sample output is provided and marked with a yellow highlighter.

Load the IMDB database

Run the script file provided: IMDB script.sql

1. Display all the rows and columns

You do not need to screenshot the entire table, just enough to show that it works.

```
1 • USE imdb;
2
3 • SELECT *
4 FROM mytable;
```

director_name	duration	director_facebook_likes	actor_3_facebook_likes	actor_2_name	actor_1_facebook_likes	gross	genres	actor_1_name	movie_title	cast_total_facebook_likes	actor_3
James Cameron	178	0	855	Joel David Moore	1000	760505847	Action Adventure Fantasy Sci-Fi	COH Pounder	Avatar	4834	Wes Stud
Gore Verbinski	169	563	1000	Orlando Bloom	40000	309404152	Action Adventure Fantasy	Johnny Depp	Pirates of the Caribbean: At World's End	48350	Jack Davi
Sam Mendes	148	0	161	Rory Kinnear	11000	200074175	Action Adventure Thriller	Christoph Waltz	Spectre	11700	Stephanie
Christopher Nolan	164	22000	22000	Christian Bale	27000	448130642	Action Thriller	Tom Hardy	The Dark Knight Rises	106759	Joseph Gr
Doug Walker	131	0	131	Rob Walker	131	0	Documentary	Doug Walker	Star Wars: Episode VII - The Force Awakens	143	0
Andrew Stanton	132	475	530	Samantha Morton	640	73058679	Action Adventure Sci-Fi	Daryl Sabara	John Carter	1873	Polly Walk
Sam Raimi	156	0	4000	James Franco	24000	336530303	Action Adventure Romance	J.K. Simmons	Spider-Man 3	46055	Kristen D.
Nathan Greno	100	15	284	Donna Murphy	799	200807262	Adventure Animation Comedy Family Fantasy ...	Brad Garrett	Tangled	2036	M.C. Gain
Joss Whedon	141	0	19000	Robert Downey Jr.	26000	458991599	Action Adventure Sci-Fi	Chris Hemsworth	Avengers: Age of Ultron	92000	Scarlett J
David Yates	153	282	10000	Daniel Radcliffe	25000	301956980	Adventure Family Fantasy Mystery	Alan Rickman	Harry Potter and the Half-Blood Prince	58753	Rupert Gr
Zack Snyder	183	0	2000	Lauren Cohan	15000	330249062	Action Adventure Sci-Fi	Henry Cavill	Batman v Superman: Dawn of Justice	24450	Frank Lan
Bryan Singer	169	0	903	Marlon Brando	18000	200069408	Action Adventure Sci-Fi	Kevin Spacey	Superman Returns	29991	Rory Kinn
Marc Forster	106	395	393	Mathieu Amalric	451	168368427	Action Adventure	Giancarlo Giannini	Quantum of Solace	2023	0
Gore Verbinski	151	563	1000	Orlando Bloom	40000	423032628	Action Adventure Fantasy	Johnny Depp	Pirates of the Caribbean: Dead Man's Chest	48486	Jack Davi
Gore Verbinski	150	563	1000	Ruth Wilson	40000	89289910	Action Adventure Western	Johnny Depp	The Lone Ranger	45757	Tom Wilks
Zack Snyder	143	0	748	Christopher Meloni	15000	291021565	Action Adventure Fantasy Sci-Fi	Henry Cavill	Man of Steel	20495	Harry Len
Andrew Adamson	150	80	201	Pierfrancesco Favino	22000	141614023	Action Adventure Family Fantasy	Peter Dinklage	The Chronicles of Narnia: Prince Caspian	22697	Damizar
Joss Whedon	173	0	19000	Robert Downey Jr.	26000	623279547	Action Adventure Sci-Fi	Chris Hemsworth	The Avengers	87697	Scarlett J
Rob Marshall	136	252	1000	Sam Claflin	40000	241063875	Action Adventure Fantasy	Johnny Depp	Pirates of the Caribbean: On Stranger Tides	54083	Stephen C
Barry Sonnenfeld	106	188	718	Michael Stuhlbarg	10000	179020854	Action Adventure Comedy Family Fantasy Sci-Fi	Will Smith	Men in Black 3	12572	Nicole Sch
Peter Jackson	164	0	773	Adam Brown	5000	255108370	Adventure Fantasy	Aidan Turner	The Hobbit: The Battle of the Five Armies	9152	James Ne
Marc Webb	153	464	963	Andrew Garfield	15000	262030663	Action Adventure Fantasy	Emma Stone	The Amazing Spider-Man	28489	Chris Zylk
Ridley Scott	156	0	738	William Hurt	891	105219735	Action Adventure Drama History	Mark Addy	Robin Hood	3244	Scott Grr
Peter Jackson	186	0	773	Adam Brown	5000	258355354	Adventure Fantasy	Aidan Turner	The Hobbit: The Desolation of Smaug	9152	James Ne
Chris Weitz	113	129	1000	Eva Green	16000	70083519	Adventure Family Fantasy	Christopher Lee	The Golden Compass	24106	Kristin Scr
Peter Jackson	201	0	84	Thomas Kretschmann	6000	718051260	Action Adventure Fantasy Romance	Nanni Moretti	Kinn Kinn	7173	Evan Dori

2. Display the `movie_title` and `title_year` for movies with a title year of 2013

```
1 • SELECT movie_title, title_year
2 FROM mytable
3 WHERE title_year = 2013;
```

Result Grid	Filter Rows:	Export:	Wrap C
movie_title	title_year		
▶ The Lone Ranger	2013		
Man of Steel	2013		
The Hobbit: The Desolation of Smaug	2013		
Iron Man 3	2013		
Monsters University	2013		
Oz the Great and Powerful	2013		
World War Z	2013		
Star Trek Into Darkness	2013		
Jack the Giant Slayer	2013		
The Great Gatsby	2013		
Pacific Rim	2013		

3. Display the movie title, duration, and genres for movies with duration of greater than 160 minutes

```
10 • SELECT movie_title, duration, genres
11 FROM mytable
12 WHERE duration > 160;
```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
movie_title	duration	genres	
▶ Avatar	178	Action Adventure Fantasy Sci-Fi	
Pirates of the Caribbean: At World's End	169	Action Adventure Fantasy	
The Dark Knight Rises	164	Action Thriller	
Batman v Superman: Dawn of Justice	183	Action Adventure Sci-Fi	
Superman Returns	169	Action Adventure Sci-Fi	
The Avengers	173	Action Adventure Sci-Fi	
The Hobbit: The Battle of the Five Armies	164	Adventure Fantasy	
The Hobbit: The Desolation of Smaug	186	Adventure Fantasy	
King Kong	201	Action Adventure Drama Romance	
Titanic	194	Drama Romance	
Iron Man 3	195	Action Adventure Sci-Fi	
Transformers: Age of Extinction	165	Action Adventure Sci-Fi	

4. Display all rows that have a title year of 2013 and are longer than 160 minutes

```
14 • SELECT
15     movie_title AS MOVIE_TITLE,
16     genres AS GENRES,
17     duration AS DURATION
18 FROM mytable
19 WHERE title_year = 2013
20     AND duration > 160;
```

Result Grid  Filter Rows: Export:  Wrap Cell Content: 			
	MOVIE_TITLE	GENRES	DURATION
▶	The Hobbit: The Desolation of Smaug	Adventure Fantasy	186
	Iron Man 3	Action Adventure Sci-Fi	195

5. Display all rows that have a null in the duration column

```
14 • SELECT
15     movie_title AS 'MOVIE_TITLE',
16     genres AS 'GENRES'
17 FROM mytable
18 WHERE duration IS NULL;
```

Result Grid  Filter Rows: Export:  Wrap Cell Content: 		
	MOVIE_TITLE	GENRES
▶	Star Wars: Episode VII - The Force Awakens	Documentary

6. Display all rows whose title begins with an A or a T and also an imdb_score greater than 8

```
20 • SELECT movie_title, imdb_score, genres
21 FROM mytable
22 WHERE movie_title REGEXP '^[AT]'
23     AND imdb_score > 8;
```

Result Grid  Filter Rows: Export:  Wrap Cell Content: 			
	movie_title	imdb_score	genres
▶	The Dark Knight Rises	8.5	Action Thriller
	The Avengers	8.1	Action Adventure Sci-Fi
	Toy Story 3	8.3	Adventure Animation Comedy Family Fantasy

7. Display all rows whose title year is 2008, 2012, or 2015 (use the IN function)
Order by year descending

```

25 • SELECT movie_title, title_year
26 FROM mytable
27 WHERE title_year IN (2008, 2012, 2015)
28 ORDER BY title_year DESC;

```

Result Grid   Filter Rows:			Export:  Wra
	movie_title	title_year	
▶	Spectre	2015	
	Avengers: Age of Ultron	2015	
	Jurassic World	2015	
	Furious 7	2015	
	The Good Dinosaur	2015	
	Jupiter Ascending	2015	
	The Dark Knight Rises	2012	
	John Carter	2012	
	The Avengers	2012	
	Men in Black 3	2012	
	The Amazing Spider-Man	2012	
	Skyfall	2012	
	Brave	2012	
	Quantum of Solace	2008	
	The Chronicles of Narni...	2008	
	Indiana Jones and the ...	2008	
	WALL•E	2008	

8. Add 20% to each of the movie durations and alias the column name as "Duration with ads"

```
30 • SELECT
31     movie_title,
32     duration,
33     duration*1.2 AS 'Duration with ads'
34 FROM mytable;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Co

	movie_title	duration	Duration with ads
▶	Avatar	178	213.6
	Pirates of the Caribbean: At World's End	169	202.8
	Spectre	148	177.6
	The Dark Knight Rises	164	196.8
	Star Wars: Episode VII - The Force Awakens	NULL	NULL
	John Carter	132	158.4
	Spider-Man 3	156	187.2
	Tangled	100	120.0
	Avengers: Age of Ultron	141	169.2
	Harry Potter and the Half-Blood Prince	153	183.6
	Batman v Superman: Dawn of Justice	183	219.6
	Superman Returns	169	202.8
	Quantum of Solace	106	127.2
	Pirates of the Caribbean: Dead Man's Chest	151	181.2
	The Lone Ranger	150	180.0
	Man of Steel	143	171.6
	The Chronicles of Narnia: Prince Caspian	150	180.0
	The Avengers	173	207.6
	Pirates of the Caribbean: On Stranger Tides	136	163.2

9. Concatenate `actor_1_name` with the text "`likes =` " and with their Facebook likes
Only select movies from 2008, 2015, 2012. Order by `title_year`, descending. Make sure the output is easy to read.

```

36 • SELECT CONCAT(actor_1_name, ' likes = ', actor_1_facebook_likes) AS 'Facebook likes'
37 FROM mytable
38 WHERE title_year IN (2008, 2015, 2012)
39 ORDER BY title_year DESC;

```

Result Grid		Filter Rows:	Export:	Wrap Cell Content:
	Facebook likes			
▶	Christoph Waltz likes = 11000			
	Chris Hemsworth likes = 26000			
	Bryce Dallas Howard likes = 3000			
	Jason Statham likes = 26000			
	A.J. Buckley likes = 275			
	Channing Tatum likes = 17000			
	Tom Hardy likes = 27000			
	Daryl Sabara likes = 640			
	Chris Hemsworth likes = 26000			
	Will Smith likes = 10000			
	Emma Stone likes = 15000			
	Albert Finney likes = 883			
	Kelly Macdonald likes = 2000			
	Giancarlo Giannini likes = 451			
	Peter Dinklage likes = 22000			
	Harrison Ford likes = 11000			
	John Ratzenberger likes = 1000			

10. Display the first 3 letters of the movie_title

```
41 • SELECT LEFT(movie_title,3) AS 'first 3 letters'
42 FROM mytable;
```

Result Grid




Filter Rows:

Export:



Wrap Cell

	first 3 letters
▶	Ava
	Pir
	Spe
	The
	Sta
	Joh
	Spi
	Tan
	Ave
	Har
	Bat
	Sup
	Qua
	Pir
	The
	Man
	The
	The

11. Round the value of each IMDB_score

```
45 • SELECT movie_title, imdb_score, ROUND(imdb_score) AS 'rounded number'
46 FROM mytable;
```

Result Grid



Filter Rows:

Export:



Wrap Cell Content:



	movie_title	imdb_score	rounded number
▶	Avatar	7.9	8
	Pirates of the Caribbean: At World's End	7.1	7
	Spectre	6.8	7
	The Dark Knight Rises	8.5	9
	Star Wars: Episode VII - The Force Awakens	7.1	7
	John Carter	6.6	7
	Spider-Man 3	6.2	6
	Tangled	7.8	8
	Avengers: Age of Ultron	7.5	8
	Harry Potter and the Half-Blood Prince	7.5	8
	Batman v Superman: Dawn of Justice	6.9	7
	Superman Returns	6.1	6
	Quantum of Solace	6.7	7
	Pirates of the Caribbean: Dead Man's Chest	7.3	7
	The Lone Ranger	6.5	7
	Man of Steel	7.2	7
	The Chronicles of Narnia: Prince Caspian	6.6	7
	The Avengers	8.1	8

12. Remove duplicate countries but do not display any **nulls**

```
49 • SELECT distinct country
50 FROM mytable
51 WHERE country IS NOT NULL;
```

Result Grid	Filter Rows:	Export:
country		
USA		
UK		
New Zealand		
Canada		
Australia		

13. Display all movies that have a IMDB score between 5 and 8
57 movies should be returned

```
54 • SELECT movie_title, imdb_score
55 FROM mytable
56 WHERE imdb_score BETWEEN 5 AND 8;
```

Result Grid	Filter Rows:	Export:	Wr
movie_title	imdb_score		
Avatar	7.9		
Pirates of the Caribbean: At World's End	7.1		
Spectre	6.8		
Star Wars: Episode VII - The Force Awakens	7.1		
John Carter	6.6		
Spider-Man 3	6.2		
Tangled	7.8		
Avengers: Age of Ultron	7.5		
Harry Potter and the Half-Blood Prince	7.5		
Batman v Superman: Dawn of Justice	6.9		
Superman Returns	6.1		
Quantum of Solace	6.7		
Pirates of the Caribbean: Dead Man's Chest	7.3		
The Lone Ranger	6.5		
Man of Steel	7.2		
The Chronicles of Narnia: Prince Caspian	6.6		
Pirates of the Caribbean: On Stranger Tides	6.7		
Men in Black 3	6.8		